Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Computer Networks & Security Mock Examination

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Total Marks: 150 Marks

Question 1 (Total: 35 marks)

A multi-site commercial business sets up a localized office infrastructure and connects it to the internet to support remote workflows.

(a) State what is meant by a **network protocol**. (1)

Protocol definition...

(b) Identify one function specific to a Network Operating System (NOS). Place a cross [X] in the correct box. (1)

[ ] **A.** Running client defragmentation loops

[ ] **B.** Separating user accounts to protect private data folders

[ ] **C.** Formatting newly installed local drive systems

[ ] **D.** Compressing raw image sets via RLE algorithms

(c) Networked devices rely on unique addresses to navigate data packets across channels.

(i) Describe the architectural structure and format of an **IPv4 address**. (2)

IPv4 layout, bits, and notation...

(ii) Describe the architectural structure and format of an **IPv6 address**. (2)

IPv6 layout, bits, and hex notation...

(iii) State two distinct methods by which an IP address can be allocated to an office workstation. (2)

Method 1

Method 2

(d) A network configuration uses a DHCP server to streamline joining nodes.
State the full meaning of the acronym **DHCP** and explain its automatic system role. (3)

Full Meaning:

Role definition:

(e) Unlike dynamic IP addresses, Media Access Control (MAC) parameters exhibit fixed properties.

(i) Identify the hardware component that carries a pre-assigned MAC identifier from the factory. (1)

Hardware component...

(ii) Describe the internal structural composition of a standard MAC address string. (2)

Composition and formatting characteristics...

(f) Explain three distinct applications or use cases for utilizing MAC addresses within wireless networks or public hotspots. (6)

Application 1

Application 2

Application 3

(g) An office worker modifies their workstation text label to "DESIGN_PC_01".
Explain why computers do not rely on device names to communicate or exchange packets across a network. (2)

Explanation...

(h) State three structural benefits provided to employees when working inside a Local Area Network (LAN). (3)

1. 2. 3.

(i) Explain two distinct administrative or backup benefits that occur when a business deploys a Client-Server network model instead of a Peer-to-Peer setup. (4)

Benefit 1

Benefit 2

(j) Identify one statement that describes a Peer-to-Peer network framework. Place a cross [X] in the correct box. (1)

[ ] **A.** Centralized backup scripts govern all client drives

[ ] **B.** Nodes share resources directly without accessing specialized servers

[ ] **C.** An authentication server distributes access token certificates

[ ] **D.** Gateway nodes rewrite packet routing protocols dynamically

(k) Define what is meant by a **roaming profile** within an automated client-server enterprise workplace. (2)

Roaming profile definition...

Question 2 (Total: 30 marks)

Wired and wireless systems depend on hardware nodes and medium selections to preserve throughput.

(a) Complete the following structural matrix evaluating specialized physical networking cables. (9)

Cable Medium Category	Maximum Data Transfer Speed	Typical Operational Context / Target Industries
Cat5e Cable	*(i)*	Homes and small business networks.
Cat6 Cable	*(ii)*	*(iii)*
Fibre Optic Cable	*(iv)*	*(v)*

(b) Explain why **fibre optic cable** paths carry signals further and faster over expansive distances than copper-based cabled wires. (2)

Explanation...

(c) Differentiate between the primary roles of a network **Switch** and a **Router**. (4)

Switch Role:

Router Role:

(d) A gateway device serves a specific purpose at the boundary of complex enterprise topologies.

(i) Define the term **gateway**. (1)

Gateway definition...

(ii) Provide an everyday practical deployment example of a gateway device inside an office network. (1)

Practical example...

(e) Standard cabled Ethernet installations and open wireless channels face physical distance constraints.

(i) State the exact maximum range limit for standard wired Ethernet cable runs before a signal drops. (1)

Maximum range...

(ii) Identify the device used to amplify network paths to extend this operational distance. (1)

Device name...

(iii) Describe how a Wireless Access Point (WAP) can be configured to achieve this extension effect. (2)

WAP configuration description...

(f) Identify one component that connects a LAN to a Wide Area Network (WAN). Place a cross [X] in the correct box. (1)

[ ] **A.** Switch Ports

[ ] **B.** Gateway Node

[ ] **C.** Booster Core

[ ] **D.** Base-16 Server

(g) Modern office spaces deploy dedicated computing towers optimized for single server tasks. Identify **three server types** and describe their specific background tasks. (6)

1. Server Type & Background Task Details:

2. Server Type & Background Task Details:

3. Server Type & Background Task Details:

Question 3 (Total: 30 marks)

Navigating internet domains requires specific software layouts, service subscriptions, and strict content checking.

(a) Accessing public digital telecommunications infrastructure requires a commercial service link.

(i) State the full meaning of the acronym **ISP**. (1)

Full meaning...

(ii) Describe how an ISP provides access paths for corporate or residential clients. (2)

Description...

(b) Educators observe that students frequently confuse Web Browsers with Search Engines.
Contrast a **web browser application** with a **search engine platform** by defining their individual functions. (4)

Web Browser Function:

Search Engine Function:

(c) School networks run filtering tools to preserve web environment safety parameters.

(i) State the full meaning of the acronym **URL**. (1)

Full meaning...

(ii) Describe the process of checking a user's web page request using **blacklist** and **whitelist** parameters. (4)

Filtration mechanism details...

(iii) State what occurs if a requested URL does not match any entry in either validation list database. (1)

System outcome...

(d) State the full meaning of the acronym **HTTP** and identify the server class that handles these requests. (2)

Full Meaning:

Server Class:

(e) Security devices preserve user data pathways against external intrusions.

(i) Identify the security device positioned at network gateways to control traffic passing to and from the internet. (1)

Device name...

(ii) State the two formats in which this security element can be deployed within an infrastructure. (2)

1. 2.

(f) Security models use mathematical encryption algorithms to alter data strings.

(i) Define the term **encryption**. (2)

Definition...

(ii) Explain why **Public Key Encryption** provides superior security over Symmetric Key Encryption. Your answer must reference key pairs, distribution risks, and decryption mechanisms. (4)

Detailed structural comparison...

(g) State three distinct methods used to secure data records across active corporate networks. (3)

1. 2. 3.

(h) Define the term **cipher**. (1)

Cipher definition...

Question 4: Technical Comparison Matrix (Total: 30 marks)

An encryption and network standard specialist is compiling a security report evaluating wireless encryption standards and remote gateway frameworks.
Complete the missing properties within the comparison blocks below by directly sourcing technical criteria from the textbook guidelines. *(30 marks, 5 marks per row/block)*

Part A: Wireless Protocols — WEP vs. WPA

Wireless Security Protocol	Key Distribution Strategy and Frequency	Overall Security Assessment Metric
Wireless Encryption Protocol (WEP)	Every connected device utilizes the exact same key string for every data transfer.	*(a)*
Wi-Fi Protected Access (WPA)	*(b)*	Highly secure alternative model.

Part B: Data Tracking, Control Records, and Remote Staging

Structural Tool Category	Core Operational Feature / Performance Criteria	Sourced Technical Use Case
Virtual Private Network (VPN)	Creates a secure connection over public networks using end-to-end data encryption.	*(c)*
File Access Rights	*(d)*	Sets read-only or read-and-write limits.
Transaction Logs	Records all system and device actions into an active history text file.	*(e)*
Geolocation Rights Management	*(f)*	Restricts user access to online tools based on geographical coordinates.

Question 5: Extended Response Essay (Total: 25 marks)

A regional health authority plans to link three medical clinics to a centralized records warehouse. The network must allow staff to access patient history files, send transactional updates, print prescription invoices locally, and connect securely when consulting from external residential sites.

The engineering architects are choosing between two structural design plans:

- **Option A:** A highly centralized Client-Server network architecture running custom Network Operating Systems. All device connections depend on fiber optic cable backbones, access control uses WPA wireless arrays coupled with MAC Address filtering rules, and all database interactions are written to automated transaction logs.
- **Option B:** A decentralized, peer-to-peer wireless architecture using standard local broadband channels. Security relies on Symmetric Key Encryption setups across the clinics, and remote consulting staff access files using software tools without centralized authentication servers.

Evaluate both options. You should discuss the impacts of each option on data confidentiality (including intercept vulnerability and access control), system scaling, data backup reliability, and structural administration overheads for the health authority's technical staff. (25)

Plan your answer carefully. Marks will be awarded for structural progression, appropriate technical ICT terminology, balanced evaluation points, and an effective final conclusion.

Write your essay response here...

End of Question Bank